

CryptoRegimeShift-LOB: A Regime-Aware L2 Crypto Benchmark for Forecast-to-Execution Degradation

Anonymous Authors

Abstract—Limit order book (LOB) forecasting models are often evaluated using directional metrics, yet such metrics can overstate deployable signal when spreads, fees, latency, visible-depth constraints, and regime shifts erode short-horizon edge. We present CryptoRegimeShift-LOB, a snapshot-level L2 crypto order book benchmark and evaluation protocol for studying forecast-to-execution degradation under microstructure regime shifts. The benchmark covers full-year BTC-USDT and ETH-USDT L2 snapshots and connects causal feature construction, cost-aware ternary labels, regime-aware forecasting, L2 snapshot replay, stress testing, bootstrap diagnostics, and BTC $\leftrightarrow$ ETH cross-asset validation. We include Robust Selective Execution Policy (RSEP) as an interpretable selective-execution diagnostic, not as a universal trading policy. The empirical evidence shows that forecasting signal can be visible while net PnL remains negative after execution frictions; RSEP mitigates losses in some evaluated settings but does not demonstrate profitability. The central finding is methodological: LOB benchmarks for financial decision contexts should be cost-aware, regime-aware, stress-tested, and explicit about the observational limits of L2 data.

Index Terms—limit order book, crypto markets, market microstructure, regime shift, forecast-to-execution degradation, selective execution, high-frequency trading benchmark

I. Introduction

Limit order book (LOB) forecasting is a core data mining task in high-frequency markets because the book records visible supply, demand, and liquidity at fine temporal resolution. Standard evaluation trains classifiers for future mid-price movement and reports accuracy, macro-F1, Matthews correlation coefficient (MCC), or balanced accuracy. These metrics measure predictive structure, but they do not establish whether a signal remains actionable after spread crossing, explicit fees, latency, limited visible depth, and regime-dependent liquidity. A forecast can be directionally correct while still failing along the path from prediction to execution.

This gap is microstructural rather than purely predictive. Trading outcomes depend on trading mechanisms, liquidity provision, adverse selection, order-book depth, and the speed at which market participants react to information [1]–[3]. In high-frequency settings, small timing changes or spread changes can alter the realized value of a short-horizon signal [4]–[6]. Crypto markets make this problem salient because trading is continuous, liquidity conditions can change sharply, and market-integrity concerns remain central [7], [8]. A benchmark for LOB models should therefore ask not only whether a model

predicts movement, but also how that signal degrades under execution frictions and changing microstructure regimes.

Recent LOB forecasting studies have substantially improved short-horizon prediction. DeepLOB and related neural approaches model spatial and temporal structure in book states [9]–[11], while benchmark and comparative studies improve experimental discipline for mid-price prediction [12]–[14]. However, prediction-centered evaluation leaves a downstream question underexamined: when forecasts are exposed to transaction costs, latency shifts, spread widening, depth shocks, and regime changes, how much of the apparent predictive edge survives?

CryptoRegimeShift-LOB addresses this question as an applied benchmark for tracing short-horizon LOB forecasts through observable execution frictions. The data are snapshot-level L2 observations rather than event-level L3 or market-by-order records, so the execution layer uses a visible-depth market-order replay approximation. This boundary keeps the benchmark focused on reproducible degradation analysis while reserving hidden liquidity, queue position, passive fills, and live venue behavior for future data settings.

The paper makes four applied benchmark contributions. First, it provides a full-year BTC-USDT and ETH-USDT L2 snapshot benchmark with chronological splits, causal features, and locked dataset statistics. Second, it turns short-horizon LOB prediction into a cost-aware evaluation problem through ternary labels, interpretable microstructure regimes, and by-regime diagnostics. Third, it provides a reproducible forecast-to-execution path: fixed predictions are passed through L2 snapshot replay, validation-tuned execution gates, fee/latency/spread/depth stress, and day-level bootstrap. Fourth, it uses RSEP and BTC $\leftrightarrow$ ETH transfer as diagnostic probes for when forecast signal survives, degrades, or fails under execution assumptions. The resulting benchmark is useful because it exposes failure modes that prediction-only LOB benchmarks hide.

This framing makes negative evidence an applied contribution. In financial machine learning, a model can appear attractive when evaluation stops at directional prediction, especially when a metric rewards frequent FLAT predictions or when minority UP/DOWN classes are hard to classify. The benchmark therefore separates three

questions: whether a model captures predictive structure, whether the predicted movement is large enough to clear local costs, and whether the resulting action survives execution stress. A result with positive gross directional movement but negative net PnL localizes forecast-to-execution degradation rather than merely lowering a leaderboard score.

The cross-asset experiments add a second applied diagnostic. Reporting BTC $\leftrightarrow$ ETH transfer by direction tests whether a source-tuned pipeline produces interpretable degradation on a related but different asset. The goal is not to average away asymmetry, but to expose how distribution shift, label balance, regime mix, and calibration affect the same evaluation path.

II. Related Work

A. LOB Forecasting and Benchmarks

LOB forecasting has been studied as short-horizon mid-price movement prediction. The FI-2010 benchmark helped standardize supervised LOB prediction [12]; DeepLOB showed that convolutional and recurrent layers can extract order-book structure [9]; and later work extended the modeling space to deep LOB learning, market-by-order data, and transformer-style architectures [10], [11], [15]. Recent benchmark and microstructural guide papers emphasize rigorous comparison and feature interpretation [13], [14]. These works motivate our forecasting baselines, but our novelty is not a new predictor. We use forecasting models as controlled inputs to an execution-aware benchmark.

A recurring lesson from this literature is that LOB models are sensitive to task definition. Horizon length, label threshold, class balance, and the number of visible levels can change both absolute performance and model ranking. This is why CryptoRegimeShift-LOB does not treat any single forecasting table as final evidence. Forecasting models are necessary to produce candidate signals, but the benchmark asks whether the same signals are robust after the cost-aware label construction, regime split, and execution replay are applied. In this sense, our use of DeepLOB-style literature is contextual: it motivates the input structure and model family, while the contribution of this paper is the evaluation path from prediction to degraded execution.

B. Market Microstructure and Execution Frictions

Market microstructure research shows that observed prices and realized trading outcomes depend on trading rules, visible and hidden liquidity, order flow, and adverse selection [1], [2], [16], [17]. HFT research further shows that latency and speed are economically meaningful frictions rather than implementation details [4], [5], [18]. For LOB forecasting, this implies that directional correctness is not enough: a signal must survive spread crossing, fee, depth, and timing risk. Because our data are L2 snapshots, we evaluate only a visible-depth market-order

approximation and explicitly avoid L3 or live-execution claims.

Crypto L2 data intensifies this distinction. Snapshot data records visible price levels and sizes, which is sufficient to construct spread, depth, imbalance, and market-order sweep approximations. It does not record the complete order lifecycle. Therefore, the benchmark is deliberately conservative: it evaluates costs observable from visible depth and stress perturbations, while treating unobserved queue priority, cancellation races, and hidden liquidity as outside scope. This makes the empirical claims narrower but more defensible.

C. Regime-Aware and Stress Evaluation

Aggregate test metrics can hide failures under distribution shift. Robustness and dataset-shift studies show that fixed test performance can degrade under shifted conditions [19]–[21]. Time-series evaluation also requires chronological validation to avoid leakage and overoptimistic estimates [22], [23]. Financial evaluation adds cost and data-snooping risks, motivating bootstrap and conservative backtest interpretation [24]–[28]. CryptoRegimeShift-LOB combines these ideas in a domain-specific path: causal L2 features $\rightarrow$ cost-aware labels $\rightarrow$ regime-aware forecasting $\rightarrow$ execution replay $\rightarrow$ stress testing and cross-asset diagnostics.

III. Data and Benchmark Construction

A. L2 Snapshot Data

The benchmark uses Binance BTC-USDT and ETH-USDT L2 order book snapshots for 2024, following the Crypto Lake data documentation for historical crypto market data [29]. Each row is a snapshot with 20 bid levels and 20 ask levels, represented by price and size fields, along with timestamp, sequence, exchange, and symbol metadata. The audit-level dataset contains 167,753,156 valid BTC-USDT snapshots and 114,416,283 valid ETH-USDT snapshots over 366 calendar days. After dropping rows that lack the required future horizon for labels, the feature/label stores contain 167,751,306 BTC-USDT rows and 114,414,433 ETH-USDT rows.

B. Audit, Causal Features, and Labels

Before feature construction, snapshots are audited for schema validity, timestamp validity, positive price and size fields, and non-crossed best quotes. These checks create a data version that can be linked to later forecasting, execution, stress, and bootstrap artifacts. The audit does not reconstruct missing message-level events; rows with invalid book states are cleaned or logged under the benchmark configuration.

The audit layer records interval statistics, spread statistics, depth statistics, duplicate timestamp counts, invalid price or size counts, and crossed-book counts. These values are not included merely as data-engineering metadata. They are part of the benchmark evidence path: when a

TABLE I
Chronological split audit for locked Stage 3 artifacts. Validation is explicit; test is reserved for final reporting.

Symbol	Train	Validation	Test	Boundary audit
BTC-USDT	100,650,783 rows; Jan 1–Aug 31	33,550,261 rows; Aug 31–Oct 28	33,550,262 rows; Oct 28–Dec 31	locked artifact: 0-row purge; 50-event horizon overlap audited
ETH-USDT	68,648,659 rows; Jan 1–Sep 9	22,882,887 rows; Sep 9–Nov 4	22,882,887 rows; Nov 4–Dec 31	locked artifact: 0-row purge; 50-event horizon overlap audited

result table reports forecast-to-execution degradation, the reader can trace it back to the audited snapshot scope and to the split used for model training and policy tuning. This is important for Applied Track evaluation because the contribution depends on repeatable data construction rather than a single isolated model run.

For each decision time t , features use only information available at or before t . Let $b_{t,\ell}^p, b_{t,\ell}^q$ be bid price and size at level ℓ , and let $a_{t,\ell}^p, a_{t,\ell}^q$ be ask price and size. The mid-price and spread are

$$m_t = \frac{a_{t,0}^p + b_{t,0}^p}{2}, \quad s_t = a_{t,0}^p - b_{t,0}^p. \quad (1)$$

The feature store contains spread, relative spread, multi-level depth, imbalance, order-flow-imbalance proxies, short-horizon returns and volatility, momentum, choppi-ness, liquidity-stress scores, and adverse-selection proxies. Rolling statistics end at t ; feature scalers and model parameters are fitted on training rows, regime quantiles are fitted on the training prefix, policy thresholds are tuned on validation rows, and test rows are not used for fitting or selection.

Labels are cost-aware ternary targets. For horizon h ,

$$r_{t,h} = \frac{m_{t+h} - m_t}{m_t + \epsilon}, \quad y_t = \begin{cases} \text{UP}, & r_{t,h} > \tau_t, \\ \text{DOWN}, & r_{t,h} < -\tau_t, \\ \text{FLAT}, & \text{otherwise,} \end{cases} \quad (2)$$

where $\epsilon = 10^{-9}$. The benchmark uses a fixed local threshold

$$\tau_t = (1 + \kappa)\rho_t + \frac{f}{10000}, \quad \rho_t = \frac{s_t}{m_t + \epsilon}. \quad (3)$$

The default benchmark constants are $h = 50$ events, $f = 1.0$ bps, and $\kappa = 0.5$, so $\tau_t = 1.5\rho_t + 0.0001$. The same constants are used for BTC-USDT and ETH-USDT. They are fixed in the benchmark configuration before any split is evaluated; they are not fitted, validation-tuned, or adjusted using test data. The inequalities are strict: equality with either boundary is assigned to FLAT.

This design makes the FLAT class meaningful: it is not a residual class for uncertain predictions, but a statement that the future move is too small to clear the fixed local cost proxy. Therefore, UP and DOWN labels represent movements that are directionally and cost-aware significant under the chosen horizon. Trade size, latency,

TABLE II
Default benchmark configuration for cost-aware labels. Label thresholds are fixed from configuration and are not selected on the test split.

Quantity	Default	Role
Horizon h	50 events	Future mid-price return horizon
Return $r_{t,h}$	$(m_{t+h} - m_t)/(m_t + \epsilon)$	Supervised target return
Relative spread ρ_t	$s_t/(m_t + \epsilon)$	Snapshot-local cost proxy
Fee f	1.0 bps	Explicit label friction proxy
Slippage multiplier κ	0.5	Spread-proportional buffer
Threshold τ_t	$(1 + \kappa)\rho_t + f/10000$	UP/DOWN significance boundary
Trade size q	Not used	Simulator-only parameter
Latency/depth sweep	Not used	Simulator-only parameters
Asset constants	None	Shared across BTC-USDT and ETH-USDT

and visible-depth sweep are deliberately excluded from τ_t ; they enter only later in the execution simulator.

The future return is used only to define the supervised target, not as a feature, regime input, normalization input, or policy decision variable. The benchmark uses chronological splits: training fits model parameters, feature scalers, and regime quantiles; validation tunes model-selection thresholds, execution thresholds, and RSEP parameters; and test is reserved for final reporting. The locked Stage 3 artifacts are row-contiguous chronological splits and the split audit documents a 50-event label-horizon overlap at train-validation and validation-test boundaries. To make subsequent reruns strictly purged, the split generator now drops the last h rows before each later split boundary.

IV. Regimes and Forecasting Baselines

A. Microstructure Regime Taxonomy

The regime taxonomy is a diagnostic lens, not an oracle market-state model. A deterministic assignment function maps causal diagnostics to regimes,

$$\begin{aligned} \rho_t &= g(\mathbf{z}_t), \quad \rho_t \in \mathcal{R}, \\ \mathbf{z}_t &= [z_t^{\text{spread}}, z_t^{\text{depth}}, z_t^{\text{imb}}, z_t^{\text{vol}}, \\ &\quad z_t^{\text{mom}}, z_t^{\text{chop}}, z_t^{\text{liq}}, z_t^{\text{adv}}]. \end{aligned} \quad (4)$$

The labels include CALM_LIQUID, CALM_ILLIQUID, VOLATILE_LIQUID, VOLATILE_ILLIQUID, LIQUIDITY_DROUGHT,

TABLE III
Regime construction inputs and evaluation role. All inputs are causal at decision time t .

Feature group	Examples	Role
Liquidity	Spread, relative spread, depth	Detect drought, stress, and liquid states
Volatility	Rolling returns, realized volatility	Detect volatile and shock-like states
Momentum	Imbalance, short-horizon pressure	Detect directional pressure
Choppiness	Flip rate, low persistence	Detect reversal-prone states
Adverse risk	Adverse-selection proxies	Flag fragile execution maps

MILD_LIQUIDITY_STRESS, MOMENTUM_TOXIC, CHOPPY_MEAN_REVERTING, SHOCK_RECOVERY, BALANCED_TRANSITION, and UNKNOWN. Quantile thresholds are fitted on the training split and applied chronologically. By-regime reporting tests whether average forecasting or execution metrics hide failures in liquidity-drought, volatile-illiquid, or momentum-toxic states.

Table IV lists the regime names used in reporting. The names are deliberately descriptive rather than causal claims about hidden states. The UNKNOWN state is retained for ambiguous observations so that the benchmark does not force uncertain snapshots into extreme categories.

The taxonomy audit reports UNKNOWN explicitly rather than hiding it: BTC-USDT has 13.19% overall UNKNOWN share with daily p90 15.42%, and ETH-USDT has 12.68% overall UNKNOWN share with daily p90 15.59%. Sensitivity is evaluated on deterministic one-million-row samples per asset. Under stricter extreme quantiles, UNKNOWN rises to 16.27% for BTC and 15.78% for ETH, with agreement to the baseline taxonomy of 86.48% and 85.75%, respectively. Under relaxed extremes, UNKNOWN falls to 10.57% and 10.18%, with agreement of 86.85% and 86.31%. These sensitivity results are diagnostic; they are not used to retune thresholds on test data.

By-regime evaluation is necessary because a model can look acceptable under an aggregate macro-F1 or MCC while concentrating errors, cost, or turnover in unfavorable states. Execution metrics are even more regime-sensitive because spread, available depth, and latency penalties are not constant across market conditions. A signal that is marginal in a calm liquid book may be dominated by adverse selection or cost in a momentum-toxic or liquidity-drought state. Thus, the taxonomy provides a bridge from prediction metrics to selective-execution diagnostics: if a policy reduces exposure in some regimes but fails in others, the mixed result is treated as evidence rather than hidden.

B. Forecasting Baselines

We use tabular and temporal baseline families: an SGD-trained linear classifier, a GPU XGBoost classifier [30], temporal convolutional networks (TCNs)

[31], and implementation-validated DeepLOB-style CNN-LSTM and lightweight LOB-Transformer baselines inspired by the LOBFrame model suite [14]. All temporal deep baselines use the same causal tensor contract, namely 100-event windows over the first 10 LOB levels, represented as 40 relative-price and log-size features. The locked full-row execution claims in this paper are made only for models with generated prediction and execution artifacts. DeepLOB-style and Transformer-style models are therefore treated as canonical deep LOB coverage in the artifact suite and as pilot/smoke baselines unless their full-row artifacts are explicitly reported.

The baselines are diagnostic controls rather than a claim of state-of-the-art architecture development. Table V shows that rankings depend on the metric. XGBoost has the highest accuracy, SGD and XGBoost have similar MCC, while the stride-1 TCN has the strongest macro-F1 among full-row comparable baselines but lower MCC. This disagreement motivates execution-aware evaluation.

Forecasting performance is reported with four metrics because the cost-aware ternary task is class-imbalanced. Accuracy rewards the majority class, macro-F1 gives equal weight to UP, FLAT, and DOWN, MCC provides a conservative correlation-style summary, and balanced accuracy averages recall across classes. None of these metrics is treated as a deployment criterion. They are used to diagnose whether predictive structure exists before the same predictions are passed through execution.

The stride-1 TCN is important negative evidence. If the benchmark reported only macro-F1, it would appear strongest among the full-row comparable models. Yet its downstream selective-execution result does not support a universal RSEP advantage over the cost-aware policy. Thus, forecasting metrics are reported as prediction diagnostics, not as deployment evidence.

The benchmark therefore treats model ranking as metric-dependent. If the paper reported only accuracy, XGBoost would be the apparent winner. If it reported only macro-F1, TCN variants would look stronger. If MCC is emphasized, the tabular models remain competitive. This disagreement is useful because it prevents a single metric from becoming a proxy for execution value. The next section uses exactly the same predictions to test whether these metric differences survive cost-aware replay.

V. Forecast-to-Execution Evaluation

A. L2 Snapshot Replay Specification

The execution layer converts predicted actions into market-order-style replay against visible L2 depth. This replay is a benchmark approximation: it does not model passive queue priority, hidden liquidity, cancellation races, venue routing, or live latency. Spread crossing is implicit because buys execute on ask levels and sells execute on bid levels.

For a nonzero signal at row i , entry occurs at $i + \ell$ where ℓ is the configured latency in event rows, and exit occurs

TABLE IV

Microstructure regime taxonomy. Regimes are diagnostic groups constructed from causal L2 features, not true latent market states. Thresholds are fitted on the training-prefix quantiles only.

Regime	Diagnostic interpretation	Causal inputs / rule source	Threshold source
LIQUIDITY_DROUGHT	Severe liquidity deterioration and spread/depth stress.	depth_drop_top10, spread_z_1m.	Train-prefix quantiles
MOMENTUM_TOXIC	Strong directional pressure with elevated adverse-selection risk.	momentum_score , ad-verse_selection_score.	Train-prefix quantiles
VOLATILE_ILLIQUID	High volatility with unfavorable spread and weak visible depth.	vol_score, rel_spread, total_depth_10.	Train-prefix quantiles
CHOPPY_MEAN_REVERTING	Reversal-prone movement where short-horizon direction is fragile.	choppiness_score, momentum_score .	Train-prefix quantiles
SHOCK_RECOVERY	Post-shock state with elevated movement but not strict drought.	vol_score, depth_drop_top10, spread_z_1m.	Train-prefix quantiles
VOLATILE_LIQUID	Elevated volatility with comparatively adequate depth and spread.	vol_score, rel_spread, total_depth_10.	Train-prefix quantiles
CALM_ILLIQUID	Low volatility with less favorable spread/depth conditions.	vol_score, rel_spread, total_depth_10.	Train-prefix quantiles
CALM_LIQUID	Low volatility, narrow spread, and sufficient visible depth.	vol_score, rel_spread, total_depth_10.	Train-prefix quantiles
MILD_LIQUIDITY_STRESS	Moderate liquidity deterioration below strict drought priority.	liquidity_score, stress_score, liquidity_drought_score.	Train-prefix quantiles
BALANCED_TRANSITION	Structured residual state that is not clearly calm or extreme.	stress_score, directional_toxicity_score, liquidity_score.	Train-prefix quantiles
UNKNOWN	Ambiguous or insufficient diagnostics after all priority rules.	Explicit fallback after causal rules.	Not redistributed

TABLE V

Held-out forecasting metrics. Prediction quality is necessary but not sufficient for execution value.

Model	Acc.	Macro-F1	MCC	Bal. Acc.
SGD	0.5589	0.4652	0.2363	0.4637
XGBoost	0.5677	0.4562	0.2364	0.4555
TCN stride-10	0.5282	0.4689	0.2275	0.4692
TCN stride-1	0.5281	0.4688	0.2274	0.4691

after the label horizon. The requested quantity at a replay snapshot is

$$q_t^* = \frac{N}{\max(m_t, \epsilon)}, \quad (5)$$

where N is the configured trade notional, m_t is the snapshot mid-price, and $\epsilon = 10^{-9}$. A market buy sweeps asks from level 0 to level 19; a market sell sweeps bids in the same level order. Under spread and depth stress, the executable prices and visible sizes are

$$p_{t,l}^{\text{buy}} = m_t + \alpha(p_{t,l}^{\text{ask}} - m_t), \quad d_{t,l}^{\text{buy}} = \beta d_{t,l}^{\text{ask}}, \quad (6)$$

$$p_{t,l}^{\text{sell}} = m_t - \alpha(m_t - p_{t,l}^{\text{bid}}), \quad d_{t,l}^{\text{sell}} = \beta d_{t,l}^{\text{bid}}, \quad (7)$$

where α is the spread multiplier and β is the depth multiplier. At each level, the fill is $x_l = \min(q_{\text{remaining}}, \max(d_{t,l}, 0))$. Nonfinite, nonpositive, or missing deeper-level price/size values are treated as unavailable depth for that level.

The execution price is the visible-depth VWAP,

$$\text{VWAP}_t = \frac{\sum_l x_l p_{t,l}}{\sum_l x_l}, \quad (8)$$

with fill ratio $\sum_l x_l / q_t^*$. If `partial_fill=true`, a trade is kept only when both entry and exit fill ratios are at least the configured `min_fill_ratio`; if `partial_fill=false`, any non-full sweep is converted to a zero fill and skipped. The

matched trade quantity is the smaller of entry and exit filled quantities.

Fees are charged on matched filled notional, not requested notional:

$$C = \frac{f}{10000} (qp_{\text{entry}} + qp_{\text{exit}}), \quad (9)$$

where f is the fee in basis points. Long gross PnL is $q(p_{\text{exit}}^{\text{sell}} - p_{\text{entry}}^{\text{buy}})$; short gross PnL is $q(p_{\text{entry}}^{\text{sell}} - p_{\text{exit}}^{\text{buy}})$; net PnL subtracts C . Missing required replay columns fail fast. Rows with nonfinite or nonpositive mid-price, invalid best prices or sizes, or crossed best books are treated as zero-fill rows. These rules make depth exhaustion, fee application, latency indexing, and partial fills reproducible from L2 snapshots.

The execution layer evaluates three decision mappings. The naive threshold policy trades when the predicted UP or DOWN probability exceeds a validation-tuned threshold. The cost-aware threshold policy estimates whether expected directional return exceeds spread, fee, and depth-slippage proxies. RSEP extends this gate with explicit latency, liquidity, adverse-selection, and regime-risk penalties. The RSEP λ coefficients are fixed benchmark constants from a singleton configuration grid, while θ is selected on validation data. The held-out test split is used only for final reporting and is not used to choose favorable execution assumptions.

B. RSEP Gate

RSEP gates a forecast-derived action by comparing forecast-implied expected edge with explicit cost and risk terms. Let μ_y denote the training-split mean future return for class $y \in \{\text{Up}, \text{Flat}, \text{Down}\}$, and let $p_t(y)$ be the forecast probability at time t . The implementation uses

$$\hat{e}_t = p_t(\text{Up})\mu_{\text{Up}} + p_t(\text{Flat})\mu_{\text{Flat}} + p_t(\text{Down})\mu_{\text{Down}}. \quad (10)$$

TABLE VI
Algorithm 1: RSEP selective-execution gate.

Input: forecast probabilities p_t , train class-return means μ , relative spread ρ_t , fee f , risk features, regime label, fixed λ , validation-selected θ .

1. Compute $\hat{e}_t = \sum_y p_t(y) \mu_y$.
2. Compute $\hat{c}_t = \rho_t + f/10000$.
3. Set $r_t^{\text{lat}} = |\text{latency_sensitivity_score}_t|$.
4. Set $r_t^{\text{liq}} = \max(\text{liquidity_drought_score}_t, 0)$.
5. Set $r_t^{\text{adv}} = \max(\text{adverse_selection_score}_t, 0)$.
6. Map regime to r_t^{reg} : liquidity drought 1.0, volatile illiquid 0.75, momentum toxic 0.5, choppy mean-reverting 0.25, otherwise 0.
7. Compute required edge $\hat{E}_t^{\text{req}} = \hat{c}_t + \lambda_{\text{lat}} r_t^{\text{lat}} + \lambda_{\text{liq}} r_t^{\text{liq}} + \lambda_{\text{adv}} r_t^{\text{adv}} + \lambda_{\text{reg}} r_t^{\text{reg}} + \theta$.
8. Return buy if $\hat{e}_t > \hat{E}_t^{\text{req}}$, sell if $\hat{e}_t < -\hat{E}_t^{\text{req}}$, otherwise abstain.

The cost term is $\hat{c}_t = \rho_t + f/10000$, where ρ_t is relative spread and f is the simulator fee in basis points. The gate trades only if

$$\hat{e}_t > \hat{c}_t + \lambda_{\text{lat}} \hat{r}_t^{\text{lat}} + \lambda_{\text{liq}} \hat{r}_t^{\text{liq}} + \lambda_{\text{adv}} \hat{r}_t^{\text{adv}} + \lambda_{\text{reg}} \hat{r}_t^{\text{reg}} + \theta. \quad (11)$$

For negative edge, the symmetric short rule is used: sell if $\hat{e}_t < -\hat{E}_t^{\text{req}}$, where $\hat{E}_t^{\text{req}}$ is the right-hand side of Eq. (11). Otherwise the policy abstains. Equality abstains because both comparisons are strict. RSEP is not a forecasting model and not a universal trading policy. It is an interpretable diagnostic for testing whether selective execution reduces exposure when estimated edge is too small relative to cost, latency risk, liquidity risk, adverse-selection risk, and regime risk.

C. Execution Results

Table VIII reports the main forecast-to-execution comparison. All reported policies remain net negative after costs. For SGD and XGBoost, RSEP is less negative than the cost-aware policy, but the improvement is relative loss mitigation. For the stride-1 TCN, cost-aware execution is better than RSEP. This directly blocks any claim that a stronger macro-F1 automatically yields better selective execution.

Two observations matter for reviewer interpretation. First, positive gross PnL does not survive costs in the reported configurations, which is precisely why prediction-only evaluation is incomplete. Second, lower turnover can reduce total losses, but lower turnover is not sufficient evidence of a good policy; it must be read with net PnL, per-trade loss, regime exposure, and day-level uncertainty. RSEP is therefore evaluated as a risk filter, not as a profit-maximizing optimizer. The paired day-level bootstrap artifact shows that RSEP minus cost-aware is positive for SGD and XGBoost, but crosses zero for TCN stride-1. These positive differences mean relative loss mitigation only; because absolute net PnL remains negative, they are not profitability evidence.

The RSEP ablation evidence reinforces the cautious interpretation. The full gate has net PnL $-122,646.73$ in the ablation setting, while removing latency, liquidity, adverse-selection, or regime components makes the result more negative. However, all ablation variants remain net

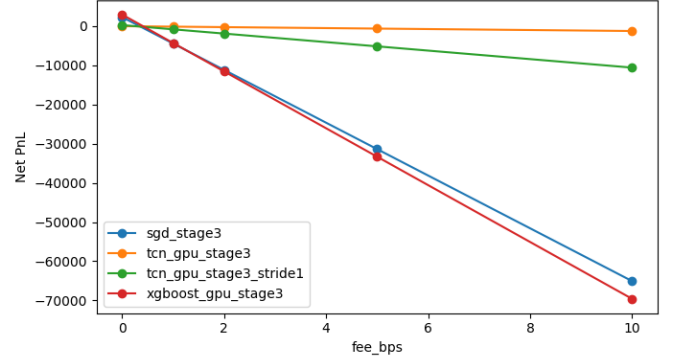


Fig. 1. Fee stress curves. The figure diagnoses transaction-cost degradation rather than profitability under favorable fees.

negative. Therefore, the evidence supports risk-filtering analysis, not a profitable policy claim.

The ablation evidence is intentionally interpreted with restraint. Removing individual risk components generally makes the result more negative, indicating that the gate filters some adverse exposures. Nevertheless, the full RSEP result remains net negative, and the stride-1 TCN result shows that RSEP is not uniformly superior to cost-aware execution. This is why the paper describes RSEP as a selective-execution diagnostic rather than as a trading policy.

VI. Stress, Robustness, and Failure Analysis

A. Stress Grid

We stress the execution layer along four axes with a fixed benchmark grid: fee $f \in \{0, 1, 2, 5, 10\}$ basis points, latency $\ell \in \{0, 1, 5, 10\}$ snapshot events, spread multiplier $\alpha_s \in \{1.0, 1.5, 2.0\}$, and visible-depth multiplier $\alpha_d \in \{1.0, 0.75, 0.5\}$. The default replay setting is $f = 1$ bps, $\ell = 1$ event, $\alpha_s = 1.0$, and $\alpha_d = 1.0$. We perturb exactly one axis at a time; all other execution assumptions remain at their default values. Stress results are diagnostic perturbations of the L2 snapshot replay approximation, not claims about live execution.

The stress grid is intentionally applied after model training and validation tuning. The held-out predictions, train-derived class-return means, selected policy thresholds, RSEP λ constants, and validation-selected RSEP θ are fixed before stress replay. We do not retrain models, recompute probabilities, or retune thresholds under any stress level. This makes the stress results closer to an out-of-distribution execution diagnostic: the forecast and decision parameters are fixed, while the execution environment changes. A robust model-policy mapping should degrade less severely when explicit cost, timing, or liquidity assumptions become less favorable.

Fee stress is the largest reported degradation axis. For SGD, net PnL changes from $-4,437.49$ at 1 bps to $-65,082.32$ at 10 bps. XGBoost changes from $-4,303.19$ to $-69,608.93$, and TCN stride-1 changes from -814.17 to

TABLE VII
Executable RSEP term mapping. The λ values are singleton benchmark constants; only θ is selected from validation.

Term	Code variable	Feature source / formula	Coefficient	Fixed or tuned
Expected edge	estimated_edge	$p_{up}\mu_{up} + p_{flat}\mu_{flat} + p_{down}\mu_{down}$; μ from train split	1	Derived, not tuned
Estimated cost	estimated_cost	$rel_spread + fee_bps/10000$	1	Fixed simulator config
Latency risk	latency_risk	$ latency_sensitivity_score $	$\lambda_{lat} = 0.25$	Fixed singleton grid
Liquidity risk	liquidity_risk	$\max(liquidity_drought_score, 0)$	$\lambda_{liq} = 0.25$	Fixed singleton grid
Adverse risk	adverse_risk	$\max(adverse_selection_score, 0)$	$\lambda_{adv} = 0.25$	Fixed singleton grid
Regime risk	regime_risk	categorical penalty map from regime label	$\lambda_{reg} = 0.15$	Fixed singleton grid
Margin	theta_edge	quantiles of positive validation margin	1	Validation-tuned

TABLE VIII
Forecast-to-execution comparison with day-level bootstrap confidence intervals on the held-out test split. Net PnL remains negative; results are degradation diagnostics, not profitability evidence.

Model	Policy	Trades	Days	Net PnL [95% CI]	Net/trade [95% CI]
SGD	Naive	32,891	65	-4,557.58 [-5,228.91, -3,835.79]	-0.1386 [-0.1465, -0.1313]
SGD	Cost-aware	67,165	65	-8,891.63 [-9,950.32, -7,735.22]	-0.1324 [-0.1373, -0.1279]
SGD	RSEP-full	33,713	65	-4,437.49 [-5,016.42, -3,821.41]	-0.1316 [-0.1361, -0.1270]
XGBoost	Naive	880,927	65	-115,238.26 [-133,871.64, -98,314.48]	-0.1308 [-0.1331, -0.1284]
XGBoost	Cost-aware	37,945	65	-4,494.39 [-5,103.67, -3,926.69]	-0.1184 [-0.1216, -0.1151]
XGBoost	RSEP-full	36,360	65	-4,303.19 [-4,843.66, -3,796.91]	-0.1183 [-0.1215, -0.1152]
TCN stride-1	Naive	19,932	65	-3,355.28 [-4,897.41, -2,009.46]	-0.1683 [-0.1880, -0.1484]
TCN stride-1	Cost-aware	4,570	65	-788.00 [-1,124.94, -500.53]	-0.1724 [-0.2159, -0.1429]
TCN stride-1	RSEP-full	5,435	65	-814.17 [-1,041.07, -609.71]	-0.1498 [-0.1605, -0.1367]

TABLE IX
RSEP ablation diagnostics. All variants remain net negative; the table analyzes risk filtering, not profitability.

Policy	Net PnL	Worst-regime return
RSEP-full	-122,646.73	-64,999.54
No latency risk	-131,038.20	-69,444.33
No liquidity risk	-192,136.81	-75,479.42
No adverse risk	-217,306.21	-70,887.92
No regime penalty	-190,460.37	-69,532.74

TABLE X
Stress-grid endpoints for selected full-row baselines. Values are net PnL; all remain negative.

Model	Fee 10	Lat. 10	Spread 2.0	Depth 0.5
SGD	-65,082.32	-6,096.92	-4,509.68	-4,484.19
XGBoost	-69,608.93	-5,761.12	-4,418.69	-4,344.95
TCN stride-1	-10,587.25	-954.59	-819.68	-818.10

TABLE XI
RobustnessAUC by stress axis. More negative values indicate stronger degradation across the stress grid.

Model	Fee	Latency	Spread	Depth
SGD	-31,390.75	-5,317.41	-4,473.58	-4,458.78
XGBoost	-33,327.97	-5,038.93	-4,360.94	-4,322.80
TCN stride-1	-5,157.76	-888.08	-816.92	-815.82

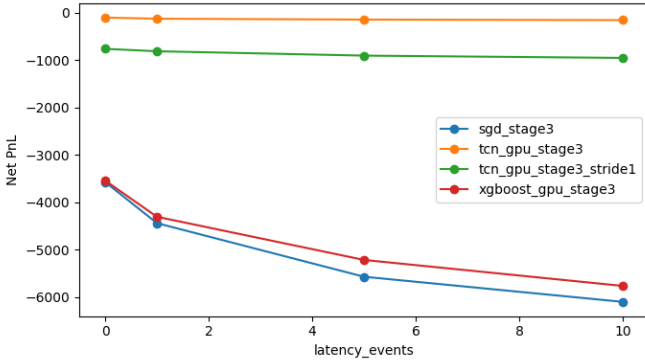


Fig. 2. Latency stress curves under the L2 snapshot replay approximation.

−10,587.25. Latency stress is negative but smaller in the locked grid: at ten latency events, SGD, XGBoost, and TCN stride-1 report −6,096.92, −5,761.12, and −954.59, respectively.

We also summarize each stress axis using a robustness

area,

$$\text{RobustnessAUC}_{m,a} = \frac{1}{|\mathcal{L}_a|} \sum_{\ell \in \mathcal{L}_a} \Pi_{m,a}^{\text{net}}(\ell), \quad (12)$$

where $\Pi_{m,a}^{\text{net}}(\ell)$ is the net PnL of model m under axis a and stress level ℓ . Because the main replay is already net negative, more negative values indicate stronger degradation across that stress axis.

The robustness table should be read together with turnover. A model that trades more frequently can accumulate larger total losses even if its per-trade loss is similar. Therefore, robustness summaries are not used to declare a universally superior model; they expose which model-policy combinations are fragile to specific execution assumptions.

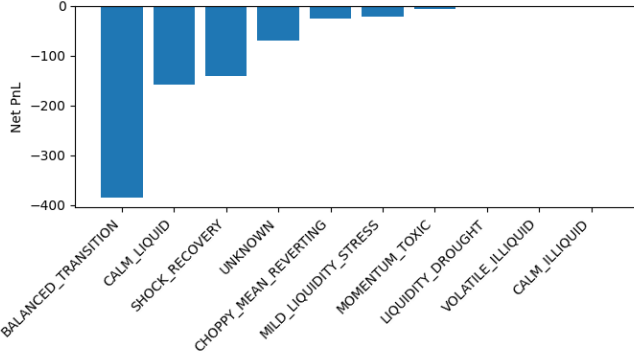


Fig. 3. Worst-regime exposure under selective-execution variants. The figure diagnoses concentration of losses; it does not imply regime-aware profitability.

B. Worst-Regime Exposure

Worst-regime diagnostics show that losses concentrate unevenly across regimes. The full RSEP gate has worst-regime return $-64,999.54$; removing liquidity risk, adverse-selection risk, or regime penalty gives $-75,479.42$, $-70,887.92$, and $-69,532.74$, respectively. These results suggest that explicit risk components reduce some adverse exposure, but do not eliminate worst-regime losses.

Failure cases include short actions in regimes such as CALM_LIQUID, MILD_LIQUIDITY_STRESS, BALANCED_TRANSITION, UNKNOWN, CHOPPY_MEAN_REVERTING, and SHOCK_RECOVERY. Thus, regime labels are useful diagnostics but not oracle safeguards. The key failure-analysis takeaway is that average forecasting quality can overstate deployable signal, and selective execution can mitigate some exposure without closing the forecast-to-execution gap.

VII. Cross-Asset Evaluation

A. Protocol and Results

The asset-held-out protocol trains and tunes on the source asset, then evaluates on the target asset test split without target validation tuning. This prevents using target-specific thresholds to make cross-asset results appear favorable. Table XII reports both directions. BTC→ETH yields macro-F1 0.4325 and MCC 0.1486; RSEP net PnL is $-74,466.38$, compared with $-287,991.44$ for cost-aware execution. ETH→BTC yields macro-F1 0.4839 and MCC 0.2424; RSEP net PnL is $-1,144.75$, compared with $-3,697.46$ for cost-aware execution.

The two directions are intentionally reported separately. A cross-asset asymmetry audit shows that the target/source median ratios differ strongly across directions: in BTC→ETH, the target ETH test split has mid-price scale $0.053\times$ the BTC source train split, relative spread $16.28\times$, top-of-book depth $5.53\times$, and total top-10 depth $7.28\times$; in ETH→BTC, the target BTC test split has mid-price scale $30.22\times$ the ETH source train split,

TABLE XII

BTC↔ETH asset-held-out evaluation. RSEP mitigates losses relative to cost-aware execution, while net PnL remains negative.

Direction	Macro-F1	MCC	RSEP net	Cost net
BTC→ETH	0.4325	0.1486	-74,466.38	-287,991.44
ETH→BTC	0.4839	0.2424	-1,144.75	-3,697.46

TABLE XIII

Cross-asset bootstrap comparison of RSEP-full minus cost-aware threshold. Positive intervals indicate relative loss mitigation only.

Direction	Mean diff.	CI low	CI high	Days
BTC→ETH	3,681.47	3,314.02	4,048.20	58
ETH→BTC	39.27	34.08	44.53	65

relative spread $0.027\times$, top-of-book depth $0.086\times$, and total top-10 depth $0.065\times$. Label and regime mix also shift: BTC→ETH evaluates on a more balanced target label mix and has target-test ECE 0.1835, while ETH→BTC evaluates on a more FLAT-heavy BTC target test split and has ECE 0.1051. These diagnostics are consistent with multi-factor distribution shift rather than a single directional cause. Averaging the two directions would hide this asymmetry and could create an overly optimistic transfer narrative. Reporting both directions preserves the failure-analysis framing: the protocol can evaluate BTC↔ETH transfer under source-validation-only tuning, but it does not imply symmetric transfer quality, universal market generalization, or profitable cross-asset trading.

Day-level bootstrap supports only a relative mitigation claim. For BTC→ETH, the mean RSEP-minus-cost-aware daily difference is 3,681.47 with CI [3,314.02, 4,048.20]. For ETH→BTC, the mean difference is 39.27 with CI [34.08, 44.53]. Both intervals are positive, but absolute net PnL remains negative. The correct generalization is narrow: BTC-USDT and ETH-USDT cross-asset forecasting and execution degradation are evaluated under this protocol; the evidence does not show market-wide transferability, profitability, or readiness for deployment.

VIII. Evidence Mapping and Reproducibility

The benchmark is organized so that each major paper claim has a direct artifact path. This is important because the empirical narrative contains both positive and negative evidence. Positive evidence includes the existence of forecasting signal, the ability to report degradation by regime, and relative loss mitigation under RSEP in selected settings. Negative evidence includes net-negative execution, stress sensitivity, and the stride-1 TCN result where stronger macro-F1 does not translate into better RSEP behavior. Without explicit evidence mapping, a reader could mistakenly interpret the paper as a trading-profit claim or, conversely, treat negative net PnL as a failed experiment. The artifact discipline prevents both errors.

The reproducibility protocol therefore separates three layers. The first layer is data construction: audit statistics,

TABLE XIV

Claim-to-evidence discipline used in the paper. The wording column states the conservative claim supported by the artifact class.

Claim class	Evidence class	artifact	Conservative wording
Benchmark scope	Dataset stats and split tables		L2 snapshot benchmark for BTC-USDT and ETH-USDT.
Regime heterogeneity	Forecasting and execution by regime		Metrics vary across microstructure regimes.
Prediction-execution gap	Forecast-to-execution tables		Forecasting gains require cost-aware execution checks.
RSEP behavior	Policy comparison and bootstrap		RSEP is a selective-execution diagnostic, not a universal winner.
Stress sensitivity	Fee/latency/spread/dept grids		Robustness is reported by stress axis, not as one PnL aggregate.
Cross-asset transfer	BTC↔ETH tables		Transfer is evaluated only for BTC-USDT and ETH-USDT.

cleaning decisions, feature/label stores, and chronological split metadata. The second layer is modeling: forecasting predictions, model labels, validation-tuned thresholds, and test-only reporting. The third layer is evaluation: execution replay outputs, stress-grid results, by-regime diagnostics, ablation tables, and bootstrap confidence intervals. Each layer is generated before the paper narrative is finalized, so the draft does not introduce numbers that cannot be traced to a canonical table or figure.

Number consistency is handled by using display values only after they are locked in result artifacts. For example, the forecasting table uses rounded display values that match the final model-selection artifact: SGD macro-F1 0.4652, XGBoost macro-F1 0.4562, and stride-1 TCN macro-F1 0.4688. The execution table uses locked policy outputs: SGD RSEP net PnL $-4,437.49$, XGBoost RSEP net PnL $-4,303.19$, and stride-1 TCN RSEP net PnL -814.17 . Cross-asset numbers are likewise reported from the asset-held-out artifacts. The manuscript avoids introducing additional unverified percentages, regime counts, or trading statistics in prose unless the same value appears in a table.

This evidence discipline also shapes the writing style. Phrases such as “survives execution”, “mitigates losses”, and “degrades under stress” are used only in relative or diagnostic form. The draft avoids language that would imply deployable profit, live-system readiness, exact market reconstruction, queue-aware simulation, or market-universal generalization. Where the results are mixed, the text says so explicitly. In particular, the paper preserves the stride-1 TCN as negative evidence because it prevents a simplistic story in which stronger forecasting metrics automatically justify selective execution. That negative evidence is valuable for ICDM-style evaluation because it exposes a failure mode of a data mining pipeline rather than hiding it behind aggregate model scores.

Finally, the benchmark records enough information for independent reruns under the same configuration:

data version, symbol, date range, feature version, split definition, model label, validation threshold, execution cost configuration, stress axis, bootstrap unit, and output artifact path. Raw commercial data may not be redistributed in the camera-ready package, but the paper can still report reproducibility artifacts, configuration files, code paths, and checksum-style summaries. This is the appropriate level of reproducibility for a large-scale financial benchmark built from licensed L2 data.

The artifact package therefore separates restricted data from public reproducibility assets. The released files include schema documentation, benchmark configs, checksum manifests, paper-ready tables, split audits, and a synthetic L2 sample with the same 20-level snapshot schema. A one-command smoke pipeline runs on this synthetic sample and exercises data audit, feature construction, cost-aware labels, regime assignment, chronological splitting, forecasting, execution replay, RSEP, stress-grid evaluation, and report generation. The synthetic sample is not used for scientific claims; it only verifies that the released code paths are executable without commercial raw data.

IX. Threats to Validity

A. Internal Validity

The main internal-validity concern is leakage. LOB data are highly autocorrelated, and even small violations of temporal order can inflate both forecasting and execution metrics. The benchmark mitigates this risk by using chronological splits, training-split normalization, validation-only policy tuning, and test-only reporting. Future mid-price values are used only to define labels. They are not used in features, regime assignments, threshold fitting, or execution decisions. The regime taxonomy is likewise causal: rolling diagnostics end at the decision time, and quantile thresholds are fitted before test evaluation. These safeguards do not prove that no implementation bug exists, but they define the intended leakage boundary and make it auditable.

A second internal-validity concern is policy overfitting. Because RSEP has several risk weights, it could be over-tuned if thresholds were selected from test results. The paper prevents this by treating validation as the only tuning split and by reporting negative cases. The stride-1 TCN is the most important check here: if the paper had tuned RSEP only to support a favorable narrative, the TCN evidence would likely be omitted. Keeping this result in the main text shows that the benchmark is not optimized to make RSEP appear universally superior.

We also maintain a model-selection ledger for the reported Stage 3 experiments. The ledger records the forecasting model family, feature and label versions, prediction artifact, threshold grid, RSEP grid, validation objective, selected thresholds, validation result, and held-out test result for each model-policy scope. Within-asset execution thresholds are selected only on chronological

validation rows; asset-held-out execution thresholds are selected only on source-asset validation rows; and stress-grid results keep predictions and selected thresholds fixed. The paper does not claim that data-snooping risk is eliminated. A formal Reality Check or PBO/CSCV diagnostic would require per-candidate daily returns for every tried configuration, whereas the current saved artifacts retain full daily/trade returns only for selected policies. We therefore report conservative controls—the full candidate ledger, validation-only selection, test-only reporting, day-level bootstrap, no retuning under stress, and retained negative evidence—rather than presenting an unsupported overfitting p -value.

B. Construct Validity

Construct validity concerns whether the benchmark measures the intended concept. The target concept is not live trading profit; it is forecast-to-execution degradation under observable L2 frictions. The L2 replay captures spread crossing, visible-depth sweep, explicit fee, latency shift, and stress perturbations. It does not capture hidden liquidity, queue position, order cancellations, market impact beyond the visible-depth approximation, or venue-specific routing. Therefore, net PnL in this paper should be read as a diagnostic quantity produced by a controlled replay approximation. It is useful for comparing degradation under the same assumptions, but it is not an estimate of live deployable profit.

The cost-aware label also affects construct validity. If the threshold is too small, the task collapses toward ordinary direction prediction and may overstate actionable signals. If it is too large, many observations become FLAT and the task may understate weak but real predictive structure. The chosen threshold is therefore part of the benchmark definition. The paper reports forecasting metrics, execution results, and stress tests together so that readers can see how this modeling choice propagates into downstream decisions.

C. External Validity

External validity is intentionally narrow. The benchmark evaluates Binance BTC-USDT and ETH-USDT in 2024. The results should not be generalized to all crypto assets, all venues, equities, futures, or foreign exchange without new experiments. Cross-asset BTC↔ETH results strengthen the benchmark relative to a single within-asset split, but they remain two directions inside a related pair of highly liquid crypto markets. Assets with lower liquidity, different fee tiers, different market-making behavior, or different exchange microstructure may produce different degradation profiles.

External validity is also limited by the snapshot frequency and the available depth levels. A higher-frequency message feed, deeper book, or market-by-order dataset could change estimated slippage, latency sensitivity, and

passive-fill conclusions. This does not invalidate the benchmark. It means that CryptoRegimeShift-LOB should be viewed as a reproducible L2 diagnostic layer. It can be extended when richer data become available, but the present claims remain limited to the L2 snapshot setting.

D. Interpretive Validity

The final threat is interpretive: readers may be tempted to treat lower losses as success or negative net PnL as failure. Both interpretations are incomplete. Lower losses under RSEP show that selective gates can filter some adverse conditions, but they do not prove profitability. Negative net PnL shows that the forecasting signal is thin after cost and stress, but it does not make the benchmark uninformative. The paper’s contribution is precisely to expose this gap. A prediction-only benchmark might stop after reporting macro-F1; CryptoRegimeShift-LOB continues until the signal is mapped through execution, stress, and cross-asset diagnostics.

X. Discussion

A. Practical Benchmark Takeaways

The empirical pattern across sections is consistent: forecasting quality is real but fragile. The models detect short-horizon structure in the L2 book, yet the signal changes character after the decision is exposed to spread crossing, fee, latency, depth, and regime stress. The practical value of the benchmark is not a single trading score. It is an audit trail from prediction to degradation. A practitioner can use the same path to ask where a model fails: before costs, after spread crossing, under latency, in a liquidity regime, or under BTC↔ETH transfer. This makes negative net PnL informative because it identifies which part of the financial ML pipeline destroys the apparent signal.

The first takeaway is that model ranking is evaluation-dependent. XGBoost has the highest accuracy, stride-1 TCN has the strongest macro-F1 among full-row comparable baselines, and the tabular models remain more competitive under MCC and execution diagnostics. No single metric orders the models consistently. This disagreement is not noise to be averaged away; it is evidence that LOB forecasting quality is multidimensional. CryptoRegimeShift-LOB lets users check whether a model that improves one prediction metric remains useful after turnover, cost, and selective-execution gates are considered.

The second takeaway is that the stress grid localizes failure modes. Fee stress dominates the reported grid, but latency, spread, and depth remain necessary axes because their importance can change under different venue assumptions or trade sizes. Reporting stress endpoints and robustness areas prevents the evaluation from depending on one favorable cost configuration. In applied financial settings, such sensitivity is often more informative than a

single backtest score because it tells users which execution assumption breaks the signal first.

The third takeaway is that selective execution can be compared reproducibly. RSEP, naive thresholds, and cost-aware thresholds are evaluated with fixed predictions, validation-only tuning, and day-level bootstrap. This makes the comparison useful even when all policies remain net negative: the benchmark measures whether a gate reduces exposure, shifts regime concentration, or fails under a stronger forecasting model. The stride-1 TCN result is important here because it prevents a simplistic narrative in which better macro-F1 automatically implies better selective execution.

Finally, cross-asset evaluation changes the evidential standard. Within-asset tests show that the pipeline works on one chronological split; BTC↔ETH tests ask whether the same source-tuned pipeline produces interpretable degradation on a related but different asset. Reporting the two directions separately exposes asymmetry in price scale, relative spread, depth, label mix, regime distribution, and calibration. This gives future users a reusable template for adding a new asset or venue: rerun the same evidence path, report the same diagnostics, and upgrade claims only when the new observational layer supports them.

XI. Limitations and Responsible Use

CryptoRegimeShift-LOB is bounded by snapshot-level L2 data. This boundary is useful for a public benchmark because visible-depth replay can be specified, tested, stressed, and reused across models. It also defines what the benchmark cannot observe: exact order arrivals, cancellations, queue priority, hidden liquidity, passive-fill dynamics, venue routing, and matching-engine details. Live deployment would require additional venue-specific evidence, including order-routing logs, measured latency, market-impact analysis, operational risk controls, and compliance review.

The forecasting baselines are a controlled diagnostic suite rather than a SOTA model-development claim. The execution results remain net negative across main reported policies, which is a core empirical finding of the benchmark: gross directional signal can be overwhelmed by spread crossing, fees, latency, visible-depth constraints, and regime-dependent risk. RSEP is useful as a diagnostic probe because it shows when selective gates reduce exposure or losses and when they fail; the stride-1 TCN is explicit negative evidence because cost-aware execution is better than RSEP and the RSEP-vs-cost-aware bootstrap interval crosses zero.

The regime taxonomy is an evaluation device rather than a true latent-state model. The cross-asset analysis is limited to BTC-USDT↔ETH-USDT and should be used as directional transfer evidence, not as a universal cross-market result. The benchmark is intended for research evaluation, robustness analysis, and risk-aware model

TABLE XV
Reproducibility checklist for the benchmark artifact pack.

Check	Expected status	Purpose
Schema and audit checks	Pass	Verify fields, timestamps, prices, sizes, spreads, and book state.
Causal feature tests	Pass	Ensure features and regimes are causal at decision time.
Chronological split lock	Pass	Prevent random splits and test-set policy tuning.
Model label registry	Pass	Link each prediction file to one model label and configuration.
Execution sanity tests	Pass	Verify that fees, spread crossing, latency, and depth changes affect replay as expected.
Stress-grid artifacts	Pass	Store fee, latency, spread, and depth perturbation outputs.
Bootstrap artifacts	Pass	Store day-level bootstrap summaries and CIs.
Claim-evidence map	Pass	Restrict wording to supported claims.

assessment. It should not be interpreted as investment advice, trading recommendation, or evidence that an automated strategy is ready for deployment. This responsible-use boundary is especially important in crypto markets, where market integrity, liquidity fragmentation, operational risk, and retail exposure remain policy concerns [8], [32].

Release and reuse should therefore emphasize reproducibility and risk assessment. The benchmark can support comparisons of forecasting models, cost-aware labels, stress-grid assumptions, selective-execution gates, and asset-transfer diagnostics. Future extensions can add L3 data, passive order simulation, venue-specific fee schedules, or additional assets by reusing the same evidence map and adding the missing observational layer before strengthening the claim.

XII. Reproducibility Checklist

Table XV summarizes the minimal reproducibility checks used when preparing the artifact pack. The checklist is included to make the benchmark auditable as an applied data mining contribution. The key point is that each number in the paper is produced after a named stage and is not manually recomputed inside the manuscript.

The checklist is not a substitute for public raw data access, but it reduces ambiguity in the paper. For a licensed financial dataset, raw snapshots may be too large or contractually restricted. The reproducibility burden therefore shifts to code paths, configuration files, split definitions, model identifiers, and paper-ready artifact tables. This is why the manuscript avoids unverifiable one-off numbers. If a statistic is not locked by a table, figure, or audit output, it is either omitted or described qualitatively.

This structure also helps future extensions. A researcher who adds L3 data, passive order simulation, another venue, or a new model family can reuse the same evidence map. The new experiment would need to add the missing observational layer rather than silently upgrading the claim. For example, a future L3 extension could claim queue-position analysis only after adding message-level order events and corresponding simulator tests. The current paper does not make that claim, and the checklist makes the limitation explicit.

XIII. Conclusion

This paper introduced CryptoRegimeShift-LOB, a regime-aware benchmark and evaluation protocol for snapshot-level L2 crypto order book forecasting. The benchmark connects causal features, cost-aware ternary labels, interpretable regimes, forecasting baselines, L2 snapshot execution replay, stress grids, bootstrap diagnostics, and BTC $\leftrightarrow$ ETH asset-held-out evaluation. The empirical evidence shows that forecasting signal and execution usefulness can diverge: a model may improve directional metrics while remaining fragile after costs, stress, regime shifts, or asset transfer. CryptoRegimeShift-LOB provides an applied evaluation path for measuring that divergence reproducibly. For high-frequency financial decision contexts, credible LOB evaluation should be cost-aware, regime-aware, stress-tested, and explicit about the observational limits of the data.

References

- [1] A. Madhavan, "Market microstructure: A survey," *Journal of Financial Markets*, vol. 3, no. 3, pp. 205–258, 2000.
- [2] M. O'Hara, "High frequency market microstructure," *Journal of Financial Economics*, vol. 116, no. 2, pp. 257–270, 2015.
- [3] M. D. Gould, M. A. Porter, S. Williams, M. McDonald, D. J. Fenn, and S. D. Howison, "Limit order books," *Quantitative Finance*, vol. 13, no. 11, pp. 1709–1742, 2013.
- [4] E. Budish, P. Cramton, and J. Shim, "The high-frequency trading arms race: Frequent batch auctions as a market design response," *The Quarterly Journal of Economics*, vol. 130, no. 4, pp. 1547–1621, 2015.
- [5] M. Aquilina, E. Budish, and P. O'Neill, "Quantifying the high-frequency trading 'arms race'," *The Quarterly Journal of Economics*, vol. 137, no. 1, pp. 493–564, 2022.
- [6] —, "Quantifying the high-frequency trading 'arms race'," Bank for International Settlements, BIS Working Paper 955, 2021. [Online]. Available: <https://www.bis.org/publ/work955.htm>
- [7] J. Almeida and T. C. Gonçalves, "Cryptocurrency market microstructure: A systematic literature review," *Annals of Operations Research*, vol. 332, no. 1, pp. 1035–1068, 2024.
- [8] International Organization of Securities Commissions, "Policy recommendations for crypto and digital asset markets: Final report," International Organization of Securities Commissions, Final Report FR11/2023, 2023. [Online]. Available: <https://www.iosco.org/library/pubdocs/pdf/IOSCOPD747.pdf>
- [9] Z. Zhang, S. Zohren, and S. Roberts, "DeepLOB: Deep convolutional neural networks for limit order books," *IEEE Transactions on Signal Processing*, vol. 67, no. 11, pp. 3001–3012, 2019.
- [10] J. A. Sirignano, "Deep learning for limit order books," *Quantitative Finance*, vol. 19, no. 4, pp. 549–570, 2019.
- [11] Z. Zhang, B. Lim, and S. Zohren, "Deep learning for market by order data," *Applied Mathematical Finance*, vol. 28, no. 1, pp. 79–95, 2021.
- [12] A. Ntakaris, M. Magris, J. Kannianen, M. Gabbouj, and A. Iosifidis, "Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods," *Journal of Forecasting*, vol. 37, no. 8, pp. 852–866, 2018.
- [13] M. Prata, G. Masi, L. Berti, V. Arrigoni, A. Coletta, I. Cannistraci, S. Vyetenko, P. Velardi, and N. Bartolini, "LOB-based deep learning models for stock price trend prediction: A benchmark study," *Artificial Intelligence Review*, vol. 57, no. 5, p. 116, 2024.
- [14] A. Briola, S. Bartolucci, and T. Aste, "Deep limit order book forecasting: A microstructural guide," *Quantitative Finance*, pp. 1–31, 2025.
- [15] Y. Xiao, C. Ventre, Y. Wang, H. Li, Y. Huan, and B. Liu, "LiT: Limit order book transformer," *Frontiers in Artificial Intelligence*, vol. 8, 2025.
- [16] R. Cont, S. Stoikov, and R. Talreja, "A stochastic model for order book dynamics," *Operations Research*, vol. 58, no. 3, pp. 549–563, 2010.
- [17] Á. Cartea, S. Jaimungal, and J. Penalva, *Algorithmic and High-Frequency Trading*. Cambridge, UK: Cambridge University Press, 2015.
- [18] M. Aquilina, G. Ibikunle, K. Rzaev, and X. Wang, "The speed premium: High-frequency trading and the cost of capital," Bank for International Settlements, BIS Working Paper 1290, 2025. [Online]. Available: <https://www.bis.org/publ/work1290.htm>
- [19] J. Quiñonero-Candela, M. Sugiyama, A. Schwaighofer, and N. D. Lawrence, Eds., *Dataset Shift in Machine Learning*. Cambridge, MA: MIT Press, 2009.
- [20] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. Dillon, B. Lakshminarayanan, and J. Snoek, "Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift," in *Advances in Neural Information Processing Systems*, vol. 32, 2019. [Online]. Available: <https://proceedings.neurips.cc/paper/2019/hash/8558cb408c1d76621371888657d2eb1d-Abstract.html>
- [21] P. W. Koh, S. Sagawa, H. Marklund, S. M. Xie, M. Zhang, A. Balsubramani, W. Hu, M. Yasunaga, R. L. Phillips, I. Gao, T. Lee, E. David, I. Stavness, W. Guo, B. A. Earnshaw, I. S. Haque, S. Beery, J. Leskovec, A. Kundahe, E. Pierson, S. Levine, C. Finn, and P. Liang, "WILDS: A benchmark of in-the-wild distribution shifts," in *Proceedings of the 38th International Conference on Machine Learning*, ser. *Proceedings of Machine Learning Research*, vol. 139, 2021, pp. 5637–5664. [Online]. Available: <https://proceedings.mlr.press/v139/koh21a.html>
- [22] C. Bergmeir, R. J. Hyndman, and B. Koo, "A note on the validity of cross-validation for evaluating autoregressive time series prediction," *Computational Statistics & Data Analysis*, vol. 120, pp. 70–83, 2018.
- [23] V. Cerqueira, L. Torgo, and I. Mozetič, "Evaluating time series forecasting models: An empirical study on performance estimation methods," *Machine Learning*, vol. 109, pp. 1997–2028, 2020.
- [24] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.
- [25] D. N. Politis and J. P. Romano, "The stationary bootstrap," *Journal of the American Statistical Association*, vol. 89, no. 428, pp. 1303–1313, 1994.
- [26] H. White, "A reality check for data snooping," *Econometrica*, vol. 68, no. 5, pp. 1097–1126, 2000.
- [27] D. H. Bailey, J. M. Borwein, M. López de Prado, and Q. J. Zhu, "The probability of backtest overfitting," *Journal of Computational Finance*, vol. 20, no. 4, pp. 39–69, 2016.
- [28] M. López de Prado, *Advances in Financial Machine Learning*. Hoboken, NJ: Wiley, 2018.
- [29] Crypto Lake, "Crypto lake data documentation," 2026, accessed 2026-06-05. [Online]. Available: <https://crypto-lake.com/data/>
- [30] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [31] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," 2018.

[32] Financial Stability Board, “Global regulatory framework for crypto-asset activities,” Financial Stability Board, Tech.

Rep., 2023. [Online]. Available: <https://www.fsb.org/2023/07/global-regulatory-framework-for-crypto-asset-activities/>